

# EXERCISE – 10

1.

```
SELECT department_id  
FROM departments
```

MINUS

```
SELECT DISTINCT department_id  
FROM employees  
WHERE job_id = 'ST_CLERK';
```

Output:

| DEPARTMENT_ID |
|---------------|
| 10            |
| 50            |
| 80            |
| 90            |

2.

```
SELECT country_id, country_name  
FROM countries
```

MINUS

```
SELECT c.country_id, c.country_name  
FROM countries c  
JOIN locations l ON c.country_id = l.country_id  
JOIN departments d ON l.location_id = d.location_id;
```

Output:

| COUNTRY_ID | COUNTRY_NAME             |
|------------|--------------------------|
| AU         | Australia                |
| CA         | Canada                   |
| SG         | Singapore                |
| UK         | United Kingdom           |
| US         | United States of America |

3.

```
SELECT job_id, department_id  
FROM employees  
WHERE department_id = 10
```

```

UNION ALL
SELECT job_id, department_id
FROM employees
WHERE department_id = 50
UNION ALL
SELECT job_id, department_id
FROM employees
WHERE department_id = 20;

```

**Output:**

| JOB_ID  | DEPARTMENT_ID |
|---------|---------------|
| IT_PROG | 10            |
| IT_PROG | 10            |
| SA_REP  | 50            |
| SA_REP  | 50            |
| SA_REP  | 50            |
| AD_VP   | 50            |
| AD_ASST | 50            |

**4.**

```

SELECT employee_id, job_id
FROM employees
INTERSECT
SELECT employee_id, job_id
FROM job_history;

```

**Output:**

| JOB_ID  | DEPARTMENT_ID |
|---------|---------------|
| IT_PROG | 10            |
| IT_PROG | 10            |
| SA_REP  | 50            |
| SA_REP  | 50            |
| SA_REP  | 50            |
| AD_VP   | 50            |

**5.**

```

SELECT last_name AS name, department_id, NULL AS department_name
FROM employees

```

**UNION**

```

SELECT NULL AS name, department_id, department_name

```

FROM departments;

Output:

| NAME   | DEPARTMENT_ID | DEPARTMENT_NAME |
|--------|---------------|-----------------|
| Austin | 50            | -               |
| Brown  | 30            | -               |
| Davies | 80            | -               |
| Davis  | 10            | -               |
| Fayad  | 50            | -               |
| Green  | 20            | -               |